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Project #22: Observation of Anti-Infective Use in Esophageal Perforations

Emma Kabalka
Henry Ford Health

Courtney Hooten
Henry Ford Health

Allison Bouwma
Henry Ford Health

Kristin Griebbe
Henry Ford Health

Carloyn Martz
Henry Ford Health

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Introduction

- Esophageal perforations (EPs) are a tear of the esophageal mucosal and muscle layers, posing infection risks, due to gastric contents in the thoracic space
- This rare disease state has many causes and has a mortality rate of 13.3%
- Diagnosis is challenging due to vague symptoms, leading to delayed treatment
- Prognosis worsening significantly after 24 hours
- Treatment involves surgery, endoscopy, and antibiotics
- There is no standardized institutional or national guidance, highlighting the need to assess current practices

Objectives

Examine characteristics of patients diagnosed with EPs

Assess empiric, definitive and outpatient anti-infective treatment of EPs

Characterize isolated pathogens from EPs

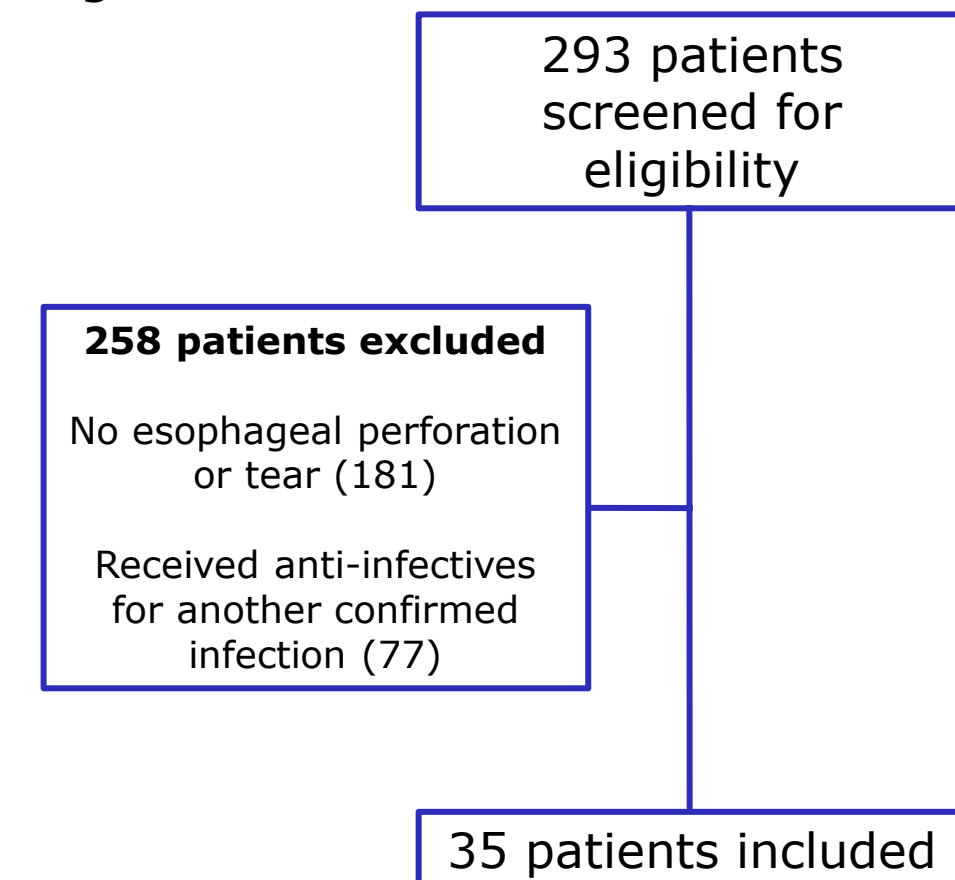
Eligibility

Inclusion	Exclusion
18 years of age or older - Documented EP or tear - Received at least one anti-infective medication after diagnosis of EP	- Vulnerable populations - Hiatal hernia without EP or tear - Received anti-infectives for another confirmed infection

Methods

- Retrospective cross-sectional study from August 1, 2021, to August 31, 2023
- The study population included patients admitted to Henry Ford Hospital, diagnosed with an esophageal perforation or tear and received at least one anti-infective agent
- Standardized forms were used for data collection
- Data was analyzed using measures of central tendency with median and interquartile range

Figure 1: Enrollment Flow Chart



Results

Table 1: Baseline Characteristics

Variable Presented as median (IQR) or n (%)	N= 35
Median age at time of admission, yrs	64 (54-72)
Female	16 (44.4)
Race	
• Black	7 (20.0)
• White	23 (65.7)
• Unknown	5 (14.2)
OSH transfer	20 (55.5)
ICU at admission	29 (82.8)
Sepsis	14 (38.9)
Septic shock	12 (33.3)
ID consulted	23 (63.9)
Death	9 (25.7)
Median LOS, days	15.5 (8-27)

Table 2: Esophageal Perforation Characteristics

Variable Presented as median (IQR) or n (%)	N=35
Cause	
• Iatrogenic	15 (42.9)
• Spontaneous	6 (17.1)
• Malignancy	5 (14.3)
• Foreign Body Ingestion	1 (2.8)
• Trauma	1 (2.8)
• Unknown	7 (20.0)
Extravasation	26 (74.3)
• Mediastinum	18 (51.4)
• Pericardial space	3 (8.6)
• Pleural space	26 (74.2)
• Empyema	14 (40.0)
• Other	1 (2.8)

Figure 2: Empiric and Definitive Coverage

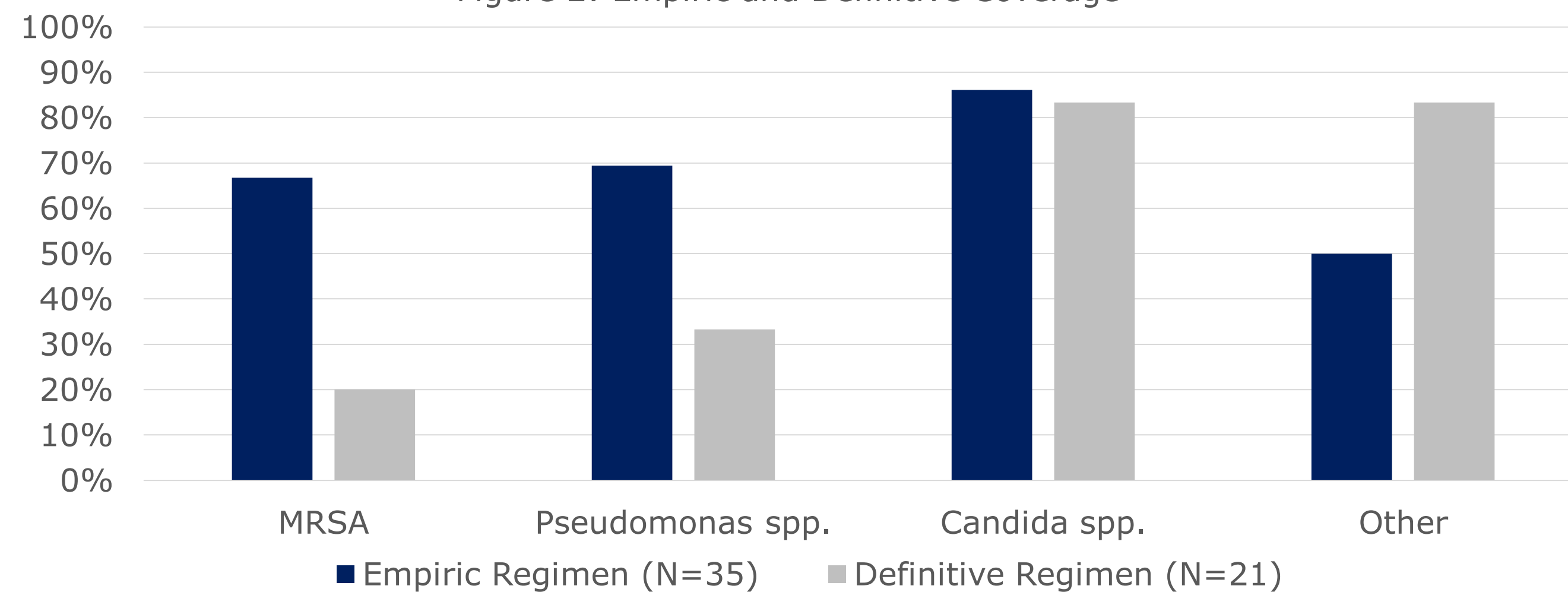


Table 3: Number of Organisms Isolated by Culture Type

Organisms isolated	Culture Type			
	Blood	Pleural fluid	Other	Total
MSSA	0	1	0	1
MRSA	0	1	0	1
<i>Staphylococcus haemolyticus</i>	1	1	0	2
<i>Streptococcus spp.</i>	0	5	1	6
<i>Enterococcus faecium</i>	0	1	0	1
<i>Enterococcus faecalis</i>	0	0	0	0
<i>Enterococcus spp. (other)</i>	0	1	0	1
<i>Klebsiella pneumonia</i>	1	2	0	3
<i>Enterobacteriaceae</i>	0	1	1	2
Anaerobes	1	2	2	5
<i>Candida</i>	0	2	2	4
Fungal (other)	0	1	1	2
No organism	14	7	8	29

Figure 4: Duration of Anti-Infective Therapy

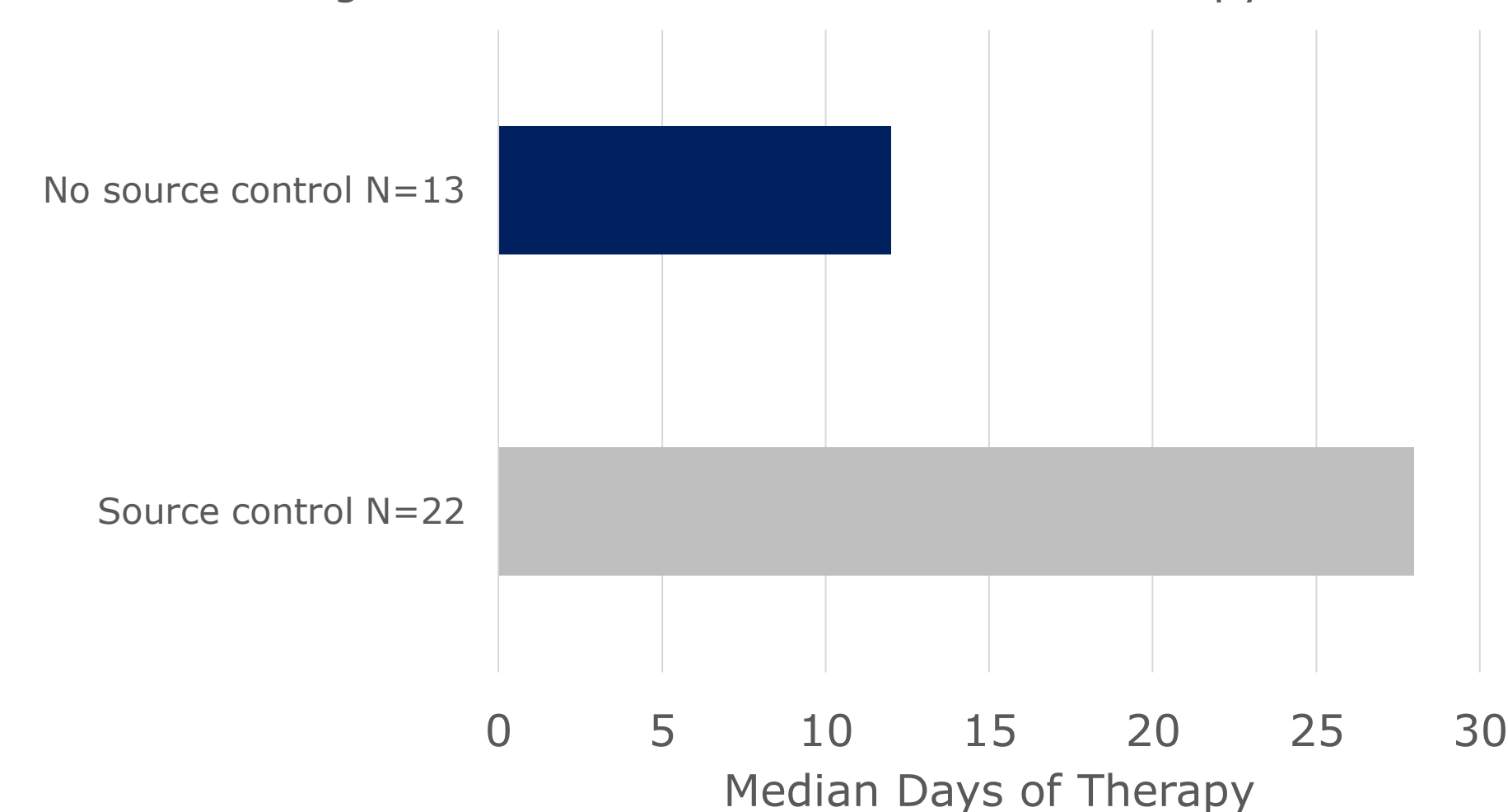


Table 4: Outpatient Anti-Infective Regimens

Variable Presented as median (IQR) or n (%)	N=18
Route of administration	
• OPAT	7 (38.9)
• Oral	11 (61.1)
Anti-infective coverage	
• MRSA and <i>Pseudomonas</i>	1 (5.6)
• MRSA only	2 (11.1)
• <i>Pseudomonas</i> only	3 (16.7)
• Other	12 (66.7)

Summary

- Most patients were white and presented from an OSH and were admitted to the ICU
- The etiology was most commonly due to iatrogenic causes
- Most common organisms isolated were MSSA, *Streptococcus* species, and fungal organisms
- Four organisms required vancomycin and/or linezolid therapy, however no *Pseudomonas* spp. were isolated
- Patients who received source control had a longer median duration of therapy than those without source control
- Of those discharged with an anti-infective regimen, the majority were on an oral regimen without MRSA and/or *Pseudomonas* coverage
- This information will be used to help develop standardized empiric antimicrobial therapy for EPs at Henry Ford Health

Key: SD: standard deviation, OSH: outside hospital, ICU: intensive care unit, ID: infectious diseases, GERD: gastroesophageal reflux disease, LOS: length of stay, MSSA: methicillin-sensitive *Staphylococcus aureus*, MRSA: methicillin-resistant *Staphylococcus aureus*, VATS: video-assisted thoracoscopy surgery